

HydroPacifique — Data Entry Reference

How each kind of measurement gets into the system: what can be imported from a CSV, what must be entered in the field, and the traps in between.

Derived from the application source, September 2026.

1. Station categories — there are exactly three

Every station belongs to one of three categories. There is no separate category for gauging stations or for well logs — those are *data recorded against* a station of one of these types.

ID	Code	Name	What the ID controls in the code
1	pluvio	Rainfall	Rain-gauge import path; statistics total with SUM instead of averaging; LAB and TOT series attach here
5	piezo	Groundwater	Benchmark and well-construction tabs on the station form; the well-log page lists only these
11	hydro	Surface water	Gauging and rating-curve pages list only these

The ID numbers are load-bearing. A category is only “Rainfall” because it is ID 1 — the behaviour is hard-coded against the literal number, not the name. A category created through Paramétrage / Types receives the next free ID (13, 14...), appears correctly in the station dropdown, and then does nothing at all: no rain-gauge handling, no totals, no well tabs. Create categories with the seeding script, not that form.

2. What can be entered, and how

Data	Initial	Station type	Entry route	Stored in
Rainfall series	RF	1 Rainfall	CSV import	chron tables
River water level	WLR	11 Surface water	CSV import	chron tables
Groundwater level	GWL	5 Groundwater	CSV import	chron tables
Stream gauging	JGE	11 Surface water	CSV import	data_jge
Rating curve (RC)	ETL	11 Surface water	CSV import	data_etl, data_etl_data
Piezo benchmarks	REP	5 Groundwater	CSV import or form	station_piezo_repere
Field report	RA	all	CSV import or form	data_ra
LAB data (rain)	LAB	1 Rainfall	CSV import	data_lab
TOT data (rain)	TOT	1 Rainfall	CSV import	data_tot
Well logs (CD) depth / conductivity / temperature	none	5 Groundwater	Field report only	data_ra_piezo_profil
Station details, well construction	—	all	Station form	station
Station photos	—	all	Station form upload	data/photos_station

Every importable type needs a matching row in `data_type` before it will be accepted — including the reserved initials. The importer validates the initial taken from the filename *before* deciding which loader to use, so a missing row fails with “No registered data series could be identified in the file name.”

3. The two CSV layouts

The initial in the filename decides which loader runs, and there are two quite different shapes.

A. Ordinary time series — RF, WLR, GWL

Three columns. No header row unless you tick the box on the import page.

```
date_heure ; valeur ; qualite  
  
13/01/2018 10:53:24 ; 1250.5 ;  
13/01/2018 11:00:00 ; 1248,2 ; 1
```

Column	Format	Notes
date_heure	dd/mm/yyyy hh:mm:ss	Several formats accepted, including yyyy-mm-dd and date-only
valeur	decimal number	Both . and , work as the decimal separator
qualite	quality code	May be left empty. If given, the code must already exist under Paramétrage / Codes qualités

B. Reserved initials — LAB, TOT, RA, JGE, ETL, REP

Each of these six is routed to its own loader with its own layout. They are **not** three-column files. JGE is the one worked out in detail so far:

Rule	Detail
Columns	25 (positions 0–24)
First two rows	Always skipped — a title line and a header line. Data starts on line 3.
Separator	Semicolon, hard-coded in the loader; it ignores the separator configured for the file type
Repeat imports	Rows update in place on (station, date/time) rather than duplicating

Positions the gauging loader actually reads:

Col	Meaning	Col	Meaning
2	Date / time (required)	14	Number of verticals
7	Mean stage H (cm)	17	Observations
8	Discharge Q (m ³ /s)	18	Agents
9	Wetted cross-sectional area (m ²)	19	GPS coordinate X
10	Mean flow velocity (m/s)	20	GPS coordinate Y
11	Surface velocity (m/s)	24	Source file name
12	Hydraulic radius (m)	—	15, 16, 21–23 are read past but not yet stored
13	Mean water depth (m)	—	

4. The exception: well logs

Well logs — the depth / conductivity / temperature profiles shown on the well-log page — are the only measurement with **no import route at all**. There is no initial for them, and the field-report import does not carry them either.

They attach to a **field report**, not directly to the station, so the path is:

1. Create the station with type **Groundwater** → **2.** Field Reports → **New FR - Groundwater** → **3.** Record the visit and enter the depth profile → **4.** The station now appears on the well-log page

The alternative route is the Nomad mobile field app, which syncs field reports and their profiles back to the server.

A new Groundwater station will not appear on the well-log page. That page lists only stations that already hold at least one profile. It is a viewing screen, not a creation screen — so an empty list means no logs have been recorded yet, not that the station is set up wrongly.

5. Two traps worth remembering

Station codes must not contain an underscore

The importer reads the station code as everything before the *first* underscore in the filename, and the series initial as what follows it.

Station code	Filename	Result
HY-001	HY-001_RF.csv	Station HY-001, series RF — works
HY_001	HY_001_RF.csv	Read as station HY, series 001 — rejected

The failure message is “No station could be identified in the file name”, which does not hint at the underscore. Use hyphens.

Adding a category through the Types screen damages the existing ones

Saving that form rewrites *every* category, not just the new one, and two settings do not survive the round trip: the point/cumulative flag and the graph style. Rainfall silently stops drawing as bars and loses its cumulative flag, with no error shown. Re-running the seeding script repairs exactly this damage.